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ANKLE

# Ability to return to sports after early lateral ligament repair of the ankle in 147 elite athletes

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## Abstract

**Purpose** The literature on elite athletes' ability to return to sports (RTS) after early lateral ligament repair of the ankle remains inadequate. The time needed to RTS after modified Broström repair for acute grade III lateral ligament injuries in a large cohort of elite athletes was evaluated while assessing the impact of other associated injuries.

**Methods** Prospective data from 147 elite athletes who underwent a primary lateral ligament repair for acute grade III injuries with clinical ankle instability from 2015 to 2019 with a minimum of 2-year follow-up were reviewed. Pertinent details such as characteristics of injury, time taken to RTS, impact of associated injuries on RTS and complications were recorded and analysed.

**Results** The average age was 24.4 years (S.D = 4.9) with 89.1% males. Approximately two-thirds of the elite athletes were football players (66.0%) and 21.1% were rugby players. Isolated lateral ligament complex injuries were sustained by 122 (83%) patients, while 25 (17%) had associated injuries (osteochondral lesions of the talus/deltoid injury/syndesmosis injury). All 147 athletes returned to their preinjury level of sports with a median time of 69 days (range 58–132 days). There was no significant difference found in time to RTS between gender, age, and types of sports. However, there was a statistically significant difference noted in the time taken to RTS earlier in patients with isolated lateral ligament injuries when compared to those with associated injuries (68.6 vs. 82.8 days; $p = 0.004$). Multiple conditional linear regression yielded three independent predictors of longer time taken to RTS which were associated deltoid injury (7.5 days longer; 95% CI 2.9–12.3; $p = 0.002$), associated syndesmosis injury (6.7 days longer; 95% CI 0.5–12.8; $p = 0.034$) and associated OLT (30.3 days longer; 95% CI 25.1–35.5; $p = 0.000$).

**Conclusion** Early surgical repair for acute grade III lateral ligament injuries in elite athletes yielded excellent rate of RTS at 9–10 weeks and a re-injury rate of 2%. Concomitant injuries will delay RTS after surgical intervention. These findings serve as important guide to managing elite athletes following acute rupture of the lateral ligament complex.

**Level of evidence** Level III.

**Keywords** Ankle lateral ligament injury · Early repair · Return to sport · Elite athletes · Broström repair

## Introduction

Ankle sprain is one of the most frequent musculoskeletal injuries sustained by athletes with an incidence of 9.35/10,000 elite athletic exposures [1]. A study reviewing a large cohort of military academy cadets reported the incidence rate of ankle sprain to be 58.4 per 1000 person-years [2]. The majority of these ankle sprains involve the lateral ligament complex with ankle supination–inversion being the most common mechanism of injury [3, 4]. Although most of the acute lateral ligament injuries respond to nonoperative management with functional rehabilitation, up to 9% of professional soccer players may sustain a re-injury the

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same season and 34% within 3 years [5, 6]. Despite controversies regarding treatment of grade III injuries, emerging evidence suggests that operative treatment for acute grade III injuries may allow earlier rehabilitation and an earlier return to sports (RTS) for elite athletes [3, 7–15]. This is because significantly less objective instability has been reported following surgical repair compared to nonoperative treatment [7, 9, 12, 13]. Since residual ankle instability is predictive of future ankle sprains [14, 16, 17], some authors have recommended considering early surgery for professional athletes with ankle instability following acute grade III lateral ligament injuries to prevent risks of complications from recurrent sprains and delayed RTS [3, 7–9, 13–15].

However, the literature on recovery and RTS following lateral ligament repair in elite athletes is sparse. Hunt et al. reviewed 360 publications describing postoperative outcomes of lateral ligament repair, but only 5.5% (20 manuscripts) provided detailed return to play timeline as an outcome metric [10]. Currently, there are only two published series with small numbers of patients (Lee et al.: 18 athletes; White et al.: 42 athletes) that have specifically reported on RTS after lateral ligament repair in elite athletes [15, 18].

The aim of this study is to evaluate the time needed to RTS following early modified Broström repair of acute grade III lateral ligament injuries in a large cohort of purely elite athletes whilst assessing the impact of other associated ankle injuries. Early lateral ligament stabilization in acute grade III injuries was hypothesized to provide a good rate of RTS for elite athletes.

## Materials and methods

Institutional (Fortius Clinic) approval to undertake this study was provided in line with the UK Health Research Authority guidance as this was considered to be an audit of clinical data and does not require ethical approval. A retrospective review of prospectively collected data was performed, and inclusion criteria included elite athletes who underwent a primary modified Broström repair by the senior author (JC) for acute grade III lateral ligament injury from 2015 to 2019 if they had a minimum of 2-year follow-up.

Those with a history of chronic instability, revision cases (re-rupture, infection, fixation failure), cases with concomitant fractures to the foot and ankle, patients with incomplete (grade I and II) injuries or those with suture-tape augmentation of the repair were excluded from this study.

Acute injuries were defined as those sustained and treated within 3 weeks of injury. Elite athletes were defined as those who were participating at national- or international-level competitions or those who were paid to perform their sporting activities. These elite athletes were indicated for surgery because of clinical and radiological diagnosis of acute grade III injuries to their lateral ligament complexes and their professional career.

Acute grade III lateral ligament injury was diagnosed by a combination of clinical and MRI evaluation. Clinical examination was performed by the senior author between 3 and 10 days following injury with instability being assessed by anterior drawer and talar tilt testing. This was the decision point for surgery unless the ankle was too swollen to make an accurate assessment in which case the patient was brought back later when swelling had reduced. The decision for surgery was made within 10 days of injury in all cases included in this study. Grade III injury was diagnosed if there was a combination of complete rupture of both the anterior talofibular ligament (ATFL) and calcaneofibular ligament (CFL) and, by comparison with the contralateral ankle, there was a grossly abnormal anterior drawer without a firm "end-point" but a sulcus sign and also significant opening on talar tilt test. This is in contrast to grade I injury which was defined as stretching of the ATFL and grade II injury being partial rupture of the ATFL with mild instability.

The medical records were reviewed for extraction of pertinent details. The demographics (age and gender), types of sports, clinical examinations, magnetic resonance imaging (MRI) findings, intraoperative findings, associated injuries, time taken to RTS, duration of follow-up, complications, re-injury and revision surgery were retrieved and analysed. Associated injuries diagnosed and their treatment modalities were also documented. RTS was defined as return to full preinjury level professional, national or international sports. The time taken to RTS was obtained from the sports club medical team and/or physiotherapists.

### Operative treatment

Surgery was performed by a single high-volume sports foot and ankle surgeon (JC). Clinical examination under anaesthesia was performed to confirm ankle laxity using anterior drawer and talar tilt tests. An anterior ankle arthroscopy was performed in all cases via standard anteromedial and anterolateral portals to allow assessment of the intra-articular structures including the cartilage surfaces, ligaments, syndesmosis stability and management of associated injuries.

A modified Broström procedure utilizing two bone anchors into the ATFL footprint of the fibula (2.4 mm × 8.5 mm Bio SutureTak from Arthrex, Naples, FL, USA). A periosteal sleeve from the fibula footprint was sutured over the top of the repair in a double breasting technique to reinforce the repair site. In this series, CFL was not routinely repaired because repair of ATFL back to the fibula footprint provided stabilization of the CFL via the arciform fibres linking the ATFL and CFL [19]. For those cases in which the ATFL was avulsed or ruptured close to

the footprint of the talus, the bone anchor was placed in the talus. Associated injuries such as an osteochondral lesion of the talus (OLT), deltoid ligament rupture (grade III injuries) and syndesmosis injuries (grade IIb injuries) and additional procedures were treated, respectively, with debridement, curettage and microfracture augmented with bone marrow aspirate concentrate for the OLT, suture anchors for repair of the deltoid ligament rupture and a suture button (Tightrope®, Arthrex, Naples, FL, USA) fixation for the syndesmosis instability.

### Postoperative rehabilitation

All patients were placed into a Plaster-of-Paris backslab (posterior splint) for 7 to 14 days, mobilized non-weight-bearing and encouraged to keep the foot elevated as much as possible to limit swelling. Sutures were removed at the 2-week wound review, and the patients were allowed incremental weight-bearing as able in a walking surgical boot (Aircast XP Walker, DJO, LLC, Vista, California) until 6 weeks. Regular physiotherapy exercises were started immediately after the removal of the backslab with focus on active ankle range of movement aiming to achieve full ankle dorsiflexion by 4 weeks. Peroneal strengthening and proprioceptive exercises commenced at 3–4 weeks postoperatively, but forced plantar flexion and hindfoot inversion were discouraged until 6 weeks. However, active plantar flexion and dorsiflexion were allowed. It is, in our experience, that plantar flexion returns very quickly and this did not prevent RTS but restricted dorsiflexion was much more likely to lead to problems returning to sports. Rehabilitation and fitness of RTS were determined by the players' club medical team and/or physiotherapists, but generally straight-line running and treadmill running were commenced after the removal of the surgical boot at 6 weeks with multidirectional training by 7–8 weeks postoperatively. There were no specific restrictions in RTS once the players had regained strength and considered fit by their medical team and/or physiotherapists although some athletes and sports clubs would use taping intermittently.

For those patients with associated OLT injuries, rehabilitation was delayed in terms of returning to impact activities such as running until 12 weeks postoperatively, while those with deltoid ligament injuries remained partial weight-bearing (50% body weight) with the use of a medial arch support from the removal of backslab to 6 weeks. Concomitant syndesmosis injuries did not alter the rehabilitation for lateral ligament repair.

### Statistical analysis

All variables are presented using frequency tables and descriptive statistics (mean, median, range and standard deviation where appropriate). Categorical variables are presented as proportions, and continuous variables as mean or median. Chi-square test and Fisher's exact test were used for comparison between categorical variables, while Mann–Whitney *U* test was used for continuous variables. Multiple linear regression models were used to control confounding factors such as age, gender and associated injuries on the time taken to RTS. Values of $p \leq 0.05$ were considered significant. In order to detect whether the study was adequately powered, post hoc sample size analysis was performed with a significance level of 0.05. Calculations performed using the difference between two means (time taken to RTS) and the number of patients in each group, respectively, showed that the current sample size was adequately powered at 92.4% to detect significant differences between the two groups [20]. Data analysis was performed using SPSS (version 16, SPSS Inc., Chicago, IL, USA).

## Results

There were 147 elite athletes with acute grade III lateral ligament injuries treated surgically with modified Broström procedure within the study period for analysis. All these athletes had confirmatory clinical findings of ankle instability consistent with MRI findings of ATFL and CFL rupture. One hundred and one patients (89.1%) were males, and the average age was 24.4 years (S.D = 4.9) for the entire cohort. Nearly two-thirds of the elite athletes were soccer players (97; 65.9%) followed by rugby players (31; 21.1%). (Table 1).

In terms of injury characteristics, 141 (95.9%) had the ATFL avulsed from the fibula footprint, while only 6 patients (4.1%) had ATFL avulsion from the talar footprint where 5 of them had isolated lateral ligament complex injuries and 1 had associated injuries. Isolated lateral ligament

**Table 1** Types of sports played by elite athletes

| Sports | Number of athletes (%) |
|---|---|
| Soccer | 97 (65.9%) |
| Rugby | 31 (21.1%) |
| Netball | 4 (2.7%) |
| Gymnastics | 4 (2.7%) |
| Cricket | 2 (1.4%) |
| Basketball | 2 (1.4%) |
| Hockey | 2 (1.4%) |
| Tennis | 2 (1.4%) |
| Athletics | 1 (0.7%) |
| Judo | 1 (0.7%) |
| Taekwondo | 1 (0.7%) |
| Total | 147 (100%) |

complex injuries were sustained by 122 (83%) patients, while 25 (17%) had associated injuries (OLT/deltoid injury/syndesmosis injury). These included 9 patients with deltoid ligament injuries, 5 patients with syndesmosis instability, 7 patients with OLT and 4 patients with combination injuries (1 patient—combined deltoid ligament and syndesmosis injuries; 2 patients—deltoid ligament injury and OLT; 1 patient—deltoid ligament, OLT and syndesmosis instability) (Table 2).

All athletes returned to their preinjury level of sports successfully with a median time taken for RTS at 69 days (mean 71 days, range 58–132 days). Athletes with isolated lateral ligament injuries alone returned at a median time of 68 days (mean 68.6 days, range 58–105 days), while those with associated injuries returned at a median of 72 days (mean 82.8 days, range 65–132 days). Three athletes (2%) re-injured the same ankle. Two soccer players reinjured the same ankle at 22 and 21 months, and one rugby player re-ruptured the same lateral ligament repair at 49 months postoperatively and required revision surgery. Four (2.7%) patients had superficial wound infection which resolved with wound dressings management and a course of oral antibiotics without any need for surgical intervention. All the 3 athletes with re-injury and 4 athletes with superficial wound infection occurred in those with isolated lateral ligament complex injuries.

When comparing time taken to RTS, there was no significant difference found between male and female (71.0 vs. 71.2 days), between team and individual sports (71.1 vs. 70.1 days) or rugby and soccer (69.3 vs. 71.8 days). However, there was a statistically significant difference noted in the time taken to RTS in patients with isolated lateral ligament injuries when compared to those with associated injuries (68.6 vs. 82.8 days; $p = 0.004$). Multiple conditional linear regression models including age, gender and types of associated injuries showed three independent predictors of longer time taken to RTS, which were associated deltoid injury (7.5 days longer; 95% CI 2.9–12.3; $p = 0.002$), associated syndesmosis injury (6.7 days longer; 95% CI 0.5–12.8; $p = 0.034$) and associated OLT (30.3 days longer; 95% CI 25.1–35.5; $p = 0.000$).

**Table 2** Types of injuries in patients with lateral ligament injury and other associated injuries

| Lateral ligament injury with other associated injuries | Number of athletes (*n*) |
|---|---|
| Deltoid ligament injuries | 9 |
| Syndesmosis instability | 5 |
| Osteochondral lesion of the talus (OLT) | 7 |
| Combined deltoid ligament and syndesmosis injuries | 1 |
| Combined deltoid ligament and OLT injuries | 2 |
| Combined deltoid ligament, OLT and syndesmosis injuries | 1 |
| Total | 25 |

## Discussion

The most important finding of this study is that modified Broström repair in elite athletes for isolated acute grade III ankle lateral ligament injuries provides a predictable return to sport within an average of 10 weeks and concomitant deltoid or syndesmosis injury may delay this return by approximately a week. This finding is particularly important because although a large proportion of acute lateral ligament injuries can heal with structured functional rehabilitation, up to 40% of such injuries may experience residual symptoms such as recurrent instability and pain during sporting activities and a significant proportion have a re-injury with subsequent missed time from training and competition [10, 16, 17, 21]. Although the re-injury rate in soccer is reported as 9% the same season and 34% over 3 years [5, 6], the re-injury rate by 2 years in this study was 2%. Furthermore, those with considerable laxity on initial presentation of a grade III injury risk failure of conservative management with functional rehabilitation and may require surgical intervention with further delay in return to competition. Therefore, surgical intervention in the form of acute lateral ligament repair may be considered in professional athletes as it provides a predictable recovery, high rate of RTS and reduced downtime [3, 7–15]. With that being said, results obtained from this cohort of elite athletes are unique and are not advocated for the general population.

This view is supported by Lee et al. who reported on the results of modified Broström repair in 18 elite athletes [18]. Surgery was performed after failure of 3 to 6 months of conservative management. The average time of return to play was 3.9 (± 1.4) months with 15 (83.3% returned by 4 months and 100% returned to preinjury level of sports by 8 months). Therefore, this delayed treatment resulted in a return to competition of 6 to 9 months following initial injury. The prolonged downtime from these injuries can significantly affect the function and livelihood of these professional athletes [7, 13, 15]. A previous series by White et al. [15] demonstrated a high RTS rate for 42 professional athletes with a median time of 77 days (range 56–178 days) after an acute grade III lateral ligament repair further reinforcing the predictable outcomes and high rates of RTS after surgical intervention for acute grade III lateral ligament injuries in elite athletes.

In a recent study on return to play for 41 patients after lateral ligament reconstruction for chronic lateral ligament instability by May et al. 32% had concomitant OLT [22]. Another study by Ferkel and Chams reported 24% prevalence of OLT in their cohort of patients with chronic lateral ligament instability [23]. Furthermore, Hinterman et al.

reported a much higher overall incidence of talar cartilage lesions at 55% in their arthroscopic review of 148 patients with chronic ankle instability [24]. They attributed cartilage damage to the repetitive ankle sprain injuries caused over time by chronic instability. These numbers are significantly higher than our study where there were only 6.8% with associated OLT. This notable difference is very likely due to the heterogeneity in patient cohort analysed and that patients in this series had acute lateral ligament injuries, whereas the earlier studies reviewed patients with chronic lateral ligament instability that had failed nonoperative treatment for at least 6 months. Chronic lateral ligament instability has been associated with development of OLT and early arthritis [2, 9, 24–27] due to altered joint biomechanics and repeated trauma to cartilage. In vivo kinematic study of the tibiotalar joints in patients with chronic lateral ankle instability showed increased anterior translation, internal rotation and superior translation of the talus [26], while in vivo peak cartilage contact strains were shown to increase in the anteromedial portion of the ankle joint [25]. Wainright et al. in a follow-up in vivo kinematic study of the tibiotalar joints in patients with chronic lateral ankle instability pre- and postoperatively reported statistically significant improvement in the abnormal ankle joint motion after a modified Broström–Gould repair which was similar to the uninjured ankle [4]. Hence, we postulate that early lateral ligament repair for acute grade III lateral ligament injuries may not only restore ankle joint stability for patients to RTS but potentially may serve to protect cartilage from repetitive injury which is pertinent for elite athletes who have a high physical demand on their ankles.

The time taken to RTS was not affected by age, gender and the type of sports (team versus individual sports and rugby versus soccer). However, associated injuries during an acute lateral ligament rupture significantly delay RTS by an average of 2 weeks when compared to isolated lateral ligament rupture alone. Using multivariate analysis to eliminate confounding effects, we found that both deltoid ligament injuries and syndesmosis injuries independently predicted an average delay to RTS by 1 week, respectively, while the presence of OLT dictated an average delay to RTS by 1 month. Although one could argue that the delay to RTS was enforced by postoperative rehabilitation rather than the injury itself, it is an important finding for the athlete and treating medical team to consider when estimating time of expected RTS. Despite this delay, our results suggest that even with associated injuries in an acute lateral ligament grade III injury, high rates of RTS can be achieved by restoring ankle joint stability through a combination of direct anatomical repair of the lateral ligaments and deltoid ligament, stabilization of the syndesmosis and bone marrow stimulation procedures for an OLT. Most importantly, these findings complement the current body of knowledge on RTS after lateral ligament injuries of the ankle in elite athletes [8, 15, 18] and provided information on the rehabilitation process and time taken to RTS for the athletes and their managing bodies, which is useful in managing expectations and potential limitations.

This study is limited by its retrospective nature of data collection, paucity of patient-based outcome measures, results from a single surgeon and the lack of a control group for comparison. There is also a possible selection bias as there was a higher prevalence of male gender in the cohort although multiple linear regression models were performed to eliminate gender as a confounding factor. Although patient-based outcome scores were not available in this cohort, reporting the ability to return to the same level of sport in elite athletes is the most important metric from an athlete and club/team's perspective. In addition, patients with associated OLT and deltoid ligament injuries had a slower rehabilitation protocol when compared to those with isolated lateral ligament injuries which could delay the time needed to RTS. Despite this limitation, the information gathered from those with associated injuries is still very useful to manage the expectations of both the athletes and their managing bodies. Besides, there are also several patients with longer time taken to RTS $\geq$ 90 days which we were not able to account for as the time taken to RTS was reported by the sports club medical team and/or physiotherapist. All patients were operated by a single surgeon which may not be generalizable, but a standardized surgical technique and postoperative rehabilitation were utilized rendering homogeneity which contributed to the robustness of the analysed results. It is also impractical to recruit a control group for comparison due to the nature of professional athletes' career and eagerness to RTS, making a randomized controlled trial almost impossible. There is a potential criticism that these athletes may have the benefit of accessing high-level rehabilitation with dedicated time to focus on RTS, but again, this is the nature of this group of elite athletes who are highly motivated with early access to high-level rehabilitative equipment and dedicated physical therapists which can positively affect the time taken to RTS. Despite these limitations, to the best of our knowledge, this study reports on the largest cohort of professional elite athletes and the time taken to RTS after surgical treatment for acute grade III lateral ligament injuries.

## Conclusion

Early surgical repair for acute grade III lateral ligament injuries in elite athletes provides a predictable RTS at an average of 10 weeks. Concomitant deltoid or syndesmosis injuries may delay this return by one week and OLT by one month. Restoration of ankle stability yielded re-injury rate of 2%

in this series. These findings serve as an important guide to athletes and their managing bodies.

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**Author contributions** Both authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by CCH and JC. The first draft of the manuscript was written by CCH, and JC commented on previous versions of the manuscript. JC supervised and reviewed the final manuscript. Both authors read and approved the final manuscript.

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**Data availability** Authors can confirm that all relevant data are included in the article and/or its supplementary information files.

## Declarations

**Conflict of interest** Choon Chiet Hong has no conflict of interest. James Calder received remuneration for speaking in educational programme by Arthrex unrelated to this study.

**Ethical approval** Institutional approval to undertake this study was provided in line with the UK Health Research Authority guidance with a waiver of consent.

**Informed consent** A waiver of consent was provided by institutional approval.

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